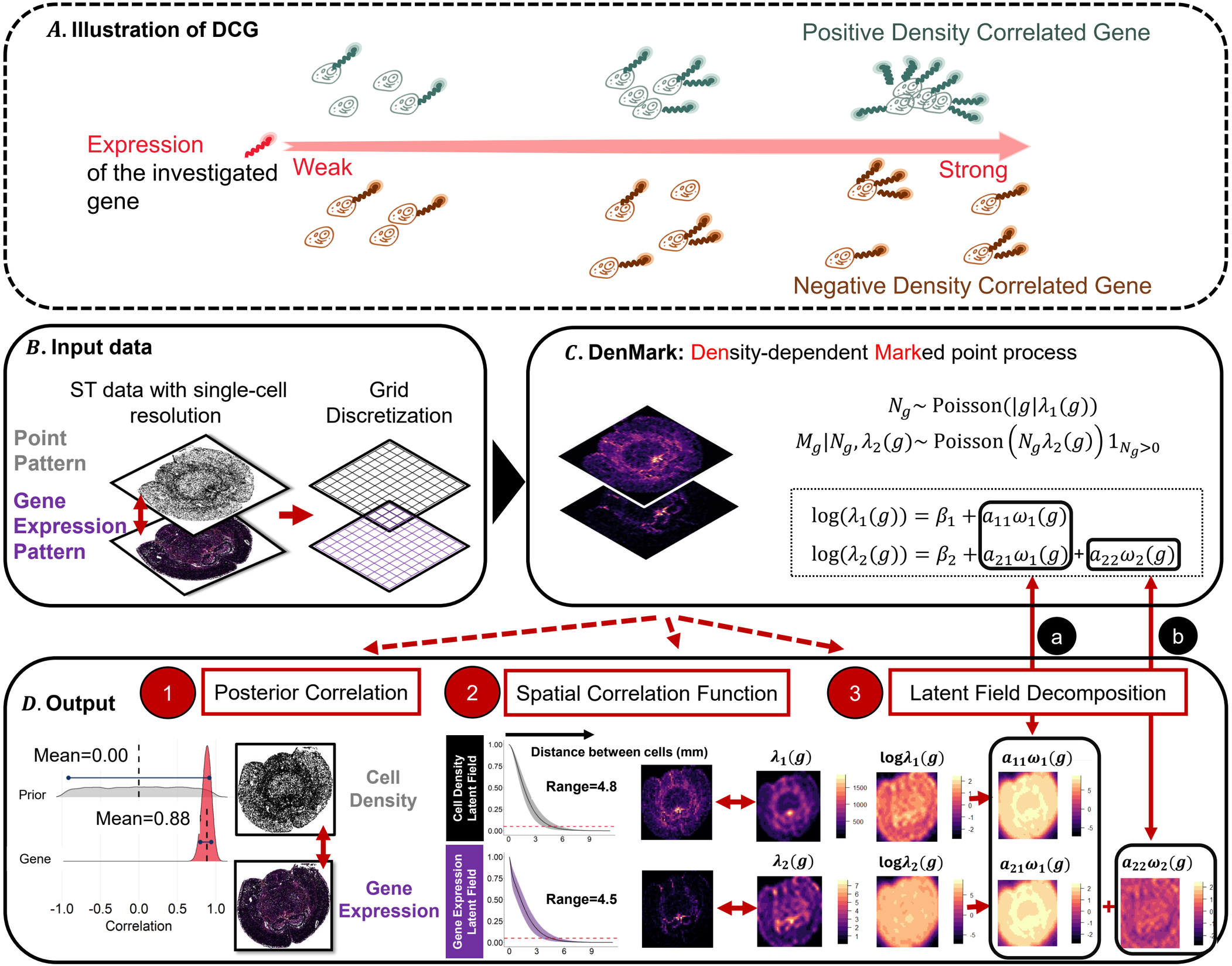
A. Illustration of DCG
Positive Density Correlated Gene
Expression of the investigated gene
Weak
Strong
Negative Density Correlated Gene
B. Input data
ST data with single-cell resolution
Grid Discretization
Point Pattern
Gene Expression Pattern
C. DenMark: Density-dependent Marked point process
$N_g \sim \text{Poisson}(|g|\lambda_1(g))$
$M_g|N_g, \lambda_2(g) \sim \text{Poisson}\left(N_g\lambda_2(g)\right)1_{N_g>0}$
$\log(\lambda_1(g)) = \beta_1 + a_{11}\omega_1(g)$
$\log(\lambda_2(g)) = \beta_2 + a_{21}\omega_1(g) + a_{22}\omega_2(g)$
a
b
D. Output
1 Posterior Correlation
2 Spatial Correlation Function
3 Latent Field Decomposition
Mean=0.00
Prior
Mean=0.88
Gene
-1.0 -0.5 0.0 0.5 1.0
Correlation
Cell Density
Gene Expression
Cell Density Latent Field
Gene Expression Latent Field
Distance between cells (mm)
Range=4.8
Range=4.5
$\lambda_1(g)$
$\log\lambda_1(g)$
$\lambda_2(g)$
$\log\lambda_2(g)$
$a_{11}\omega_1(g)$
$a_{21}\omega_1(g)$
$a_{22}\omega_2(g)$